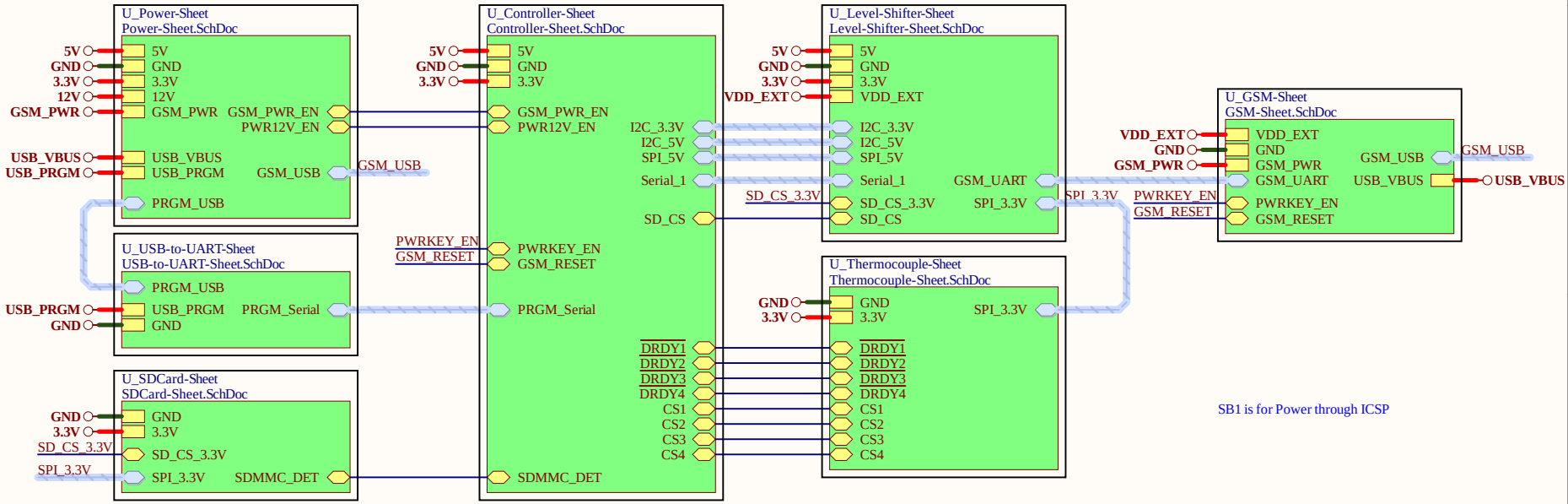
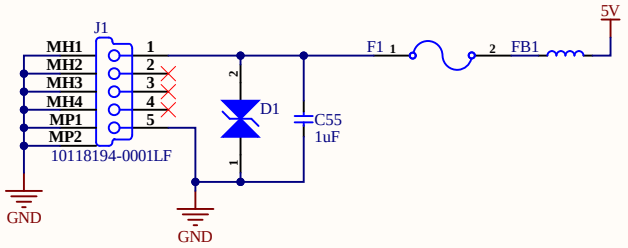


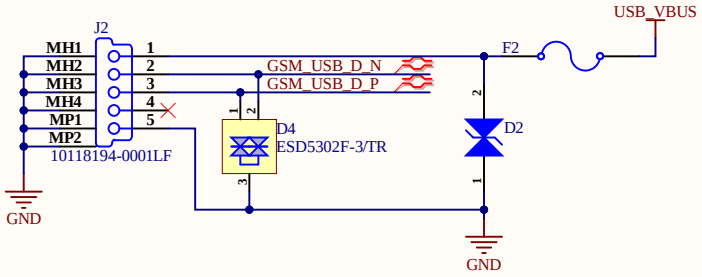
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Date: 11/1/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8

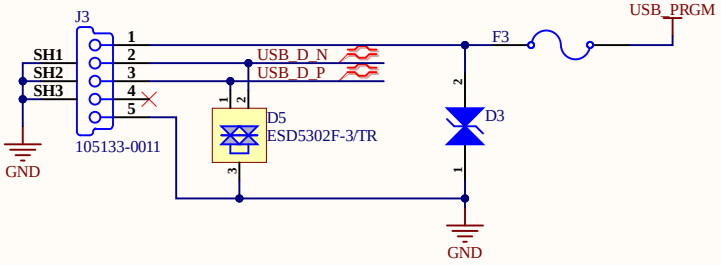
USB FOR BOARD POWER



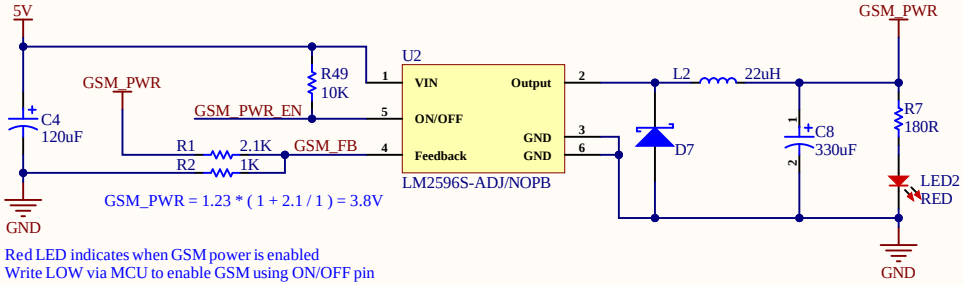
USB FOR GSM UPDATES



USB PROGRAMMING MEGA2560

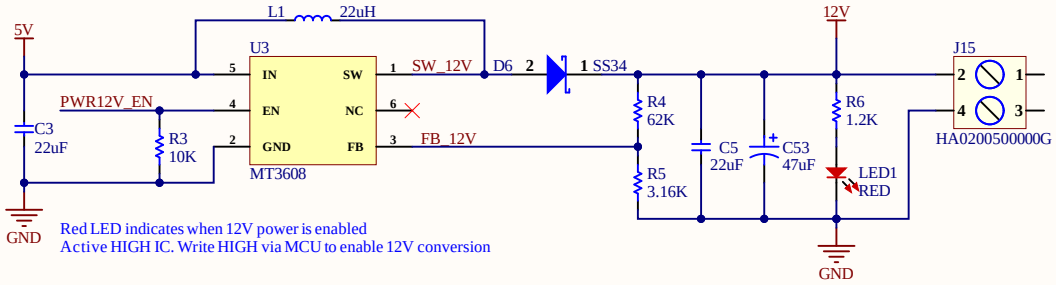


BUCK FOR GSM POWER 5V TO 3.8V



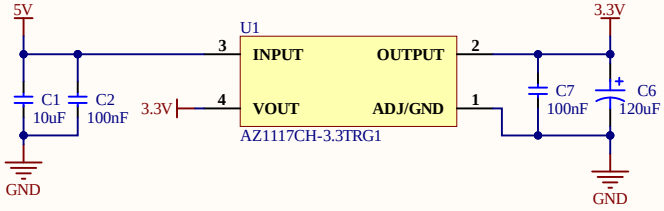
Red LED indicates when GSM power is enabled
Write LOW via MCU to enable GSM using ON/OFF pin

BOOST FOR WIKA SENSOR 5V TO 12V

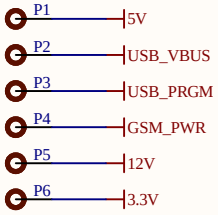


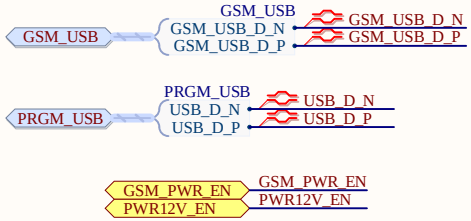
Red LED indicates when 12V power is enabled
Active HIGH IC. Write HIGH via MCU to enable 12V conversion

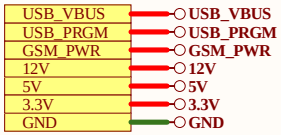
LDO 5V TO 3.3V



Test Points



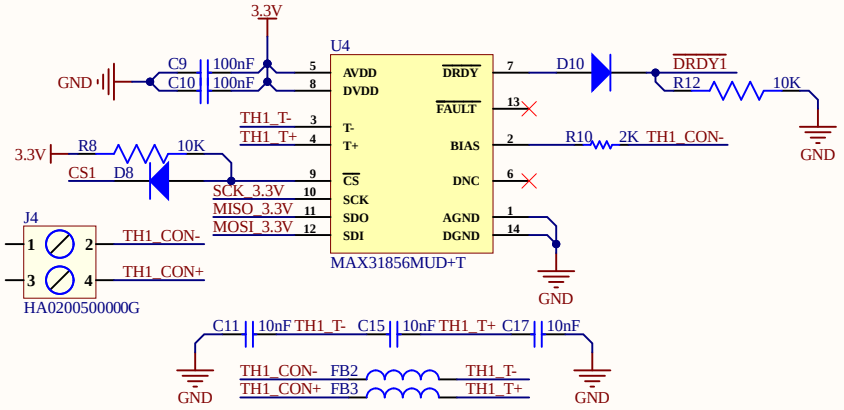




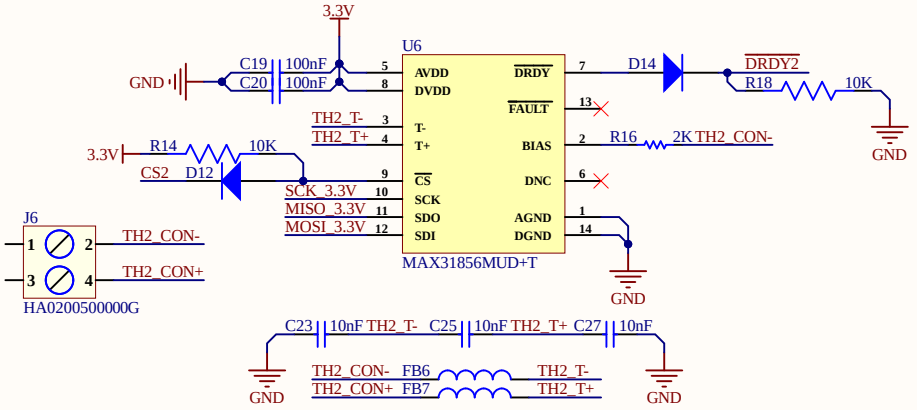
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Date: 11/1/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 8

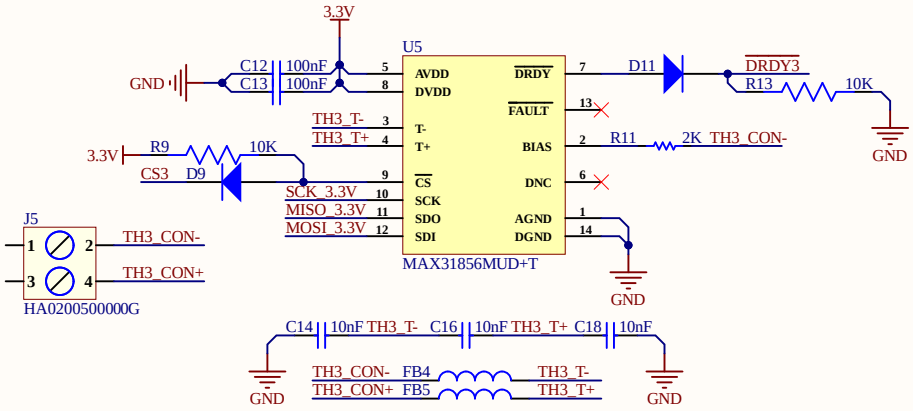
Thermocouple Input - 1



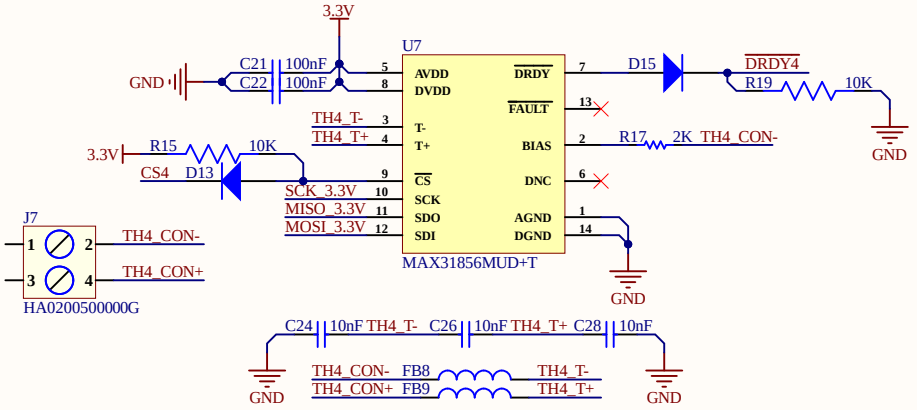
Thermocouple Input - 2



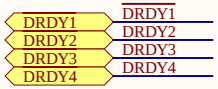
Thermocouple Input - 3

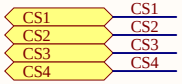


Thermocouple Input - 4







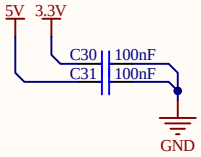


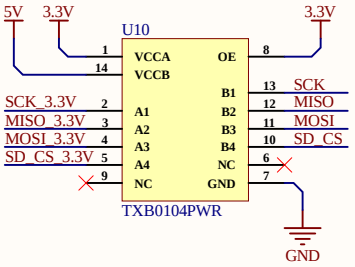


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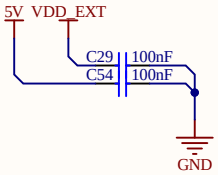
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Date: 11/1/2023	Page Num: 3
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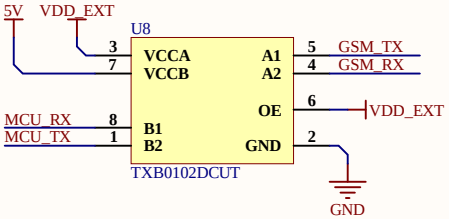
Level Shifter for SPI



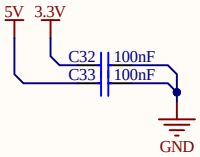


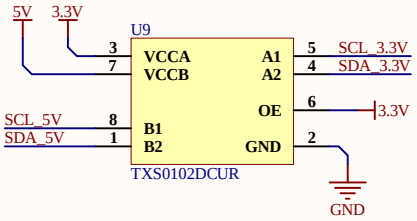
Level Shifter for Serial

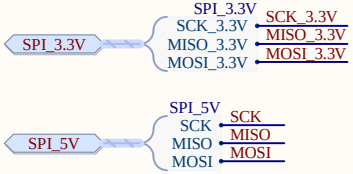


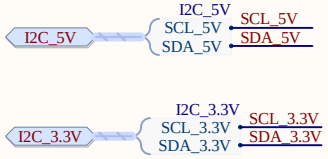


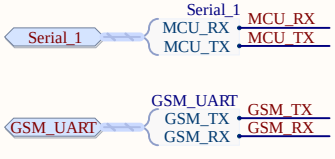
Level Shifter for I2C

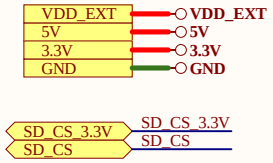






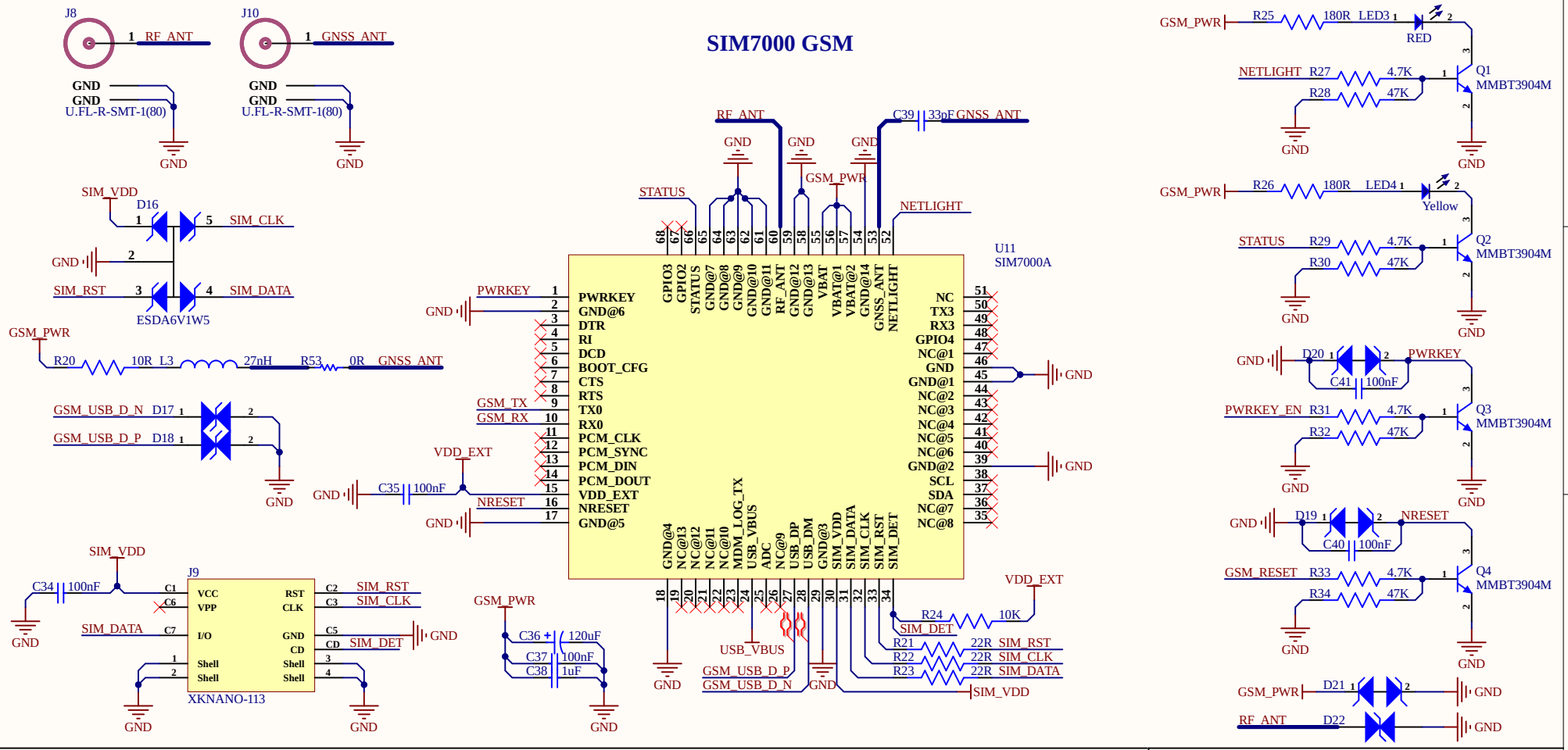






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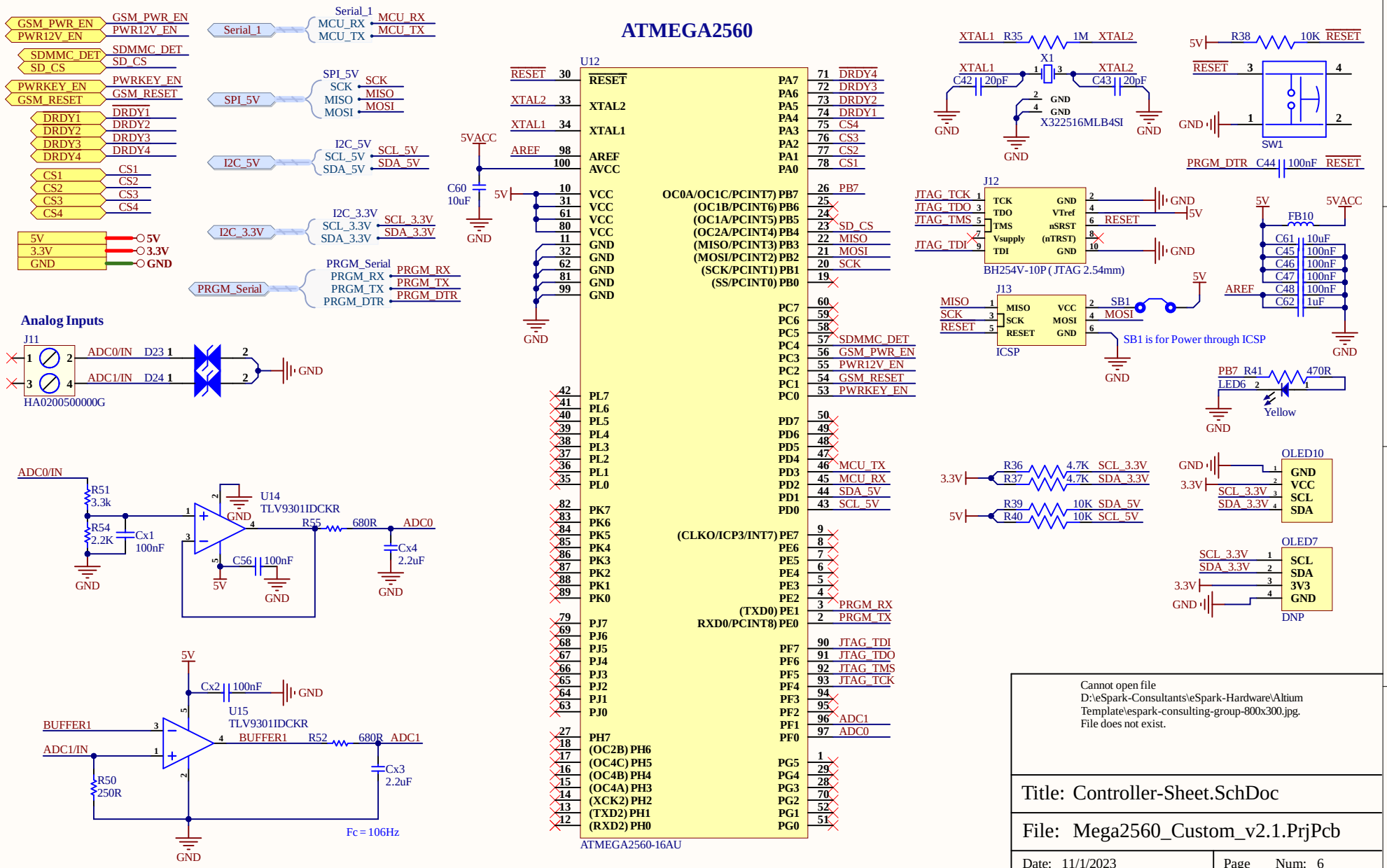
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Date: 11/1/2023	Page	Num: 4
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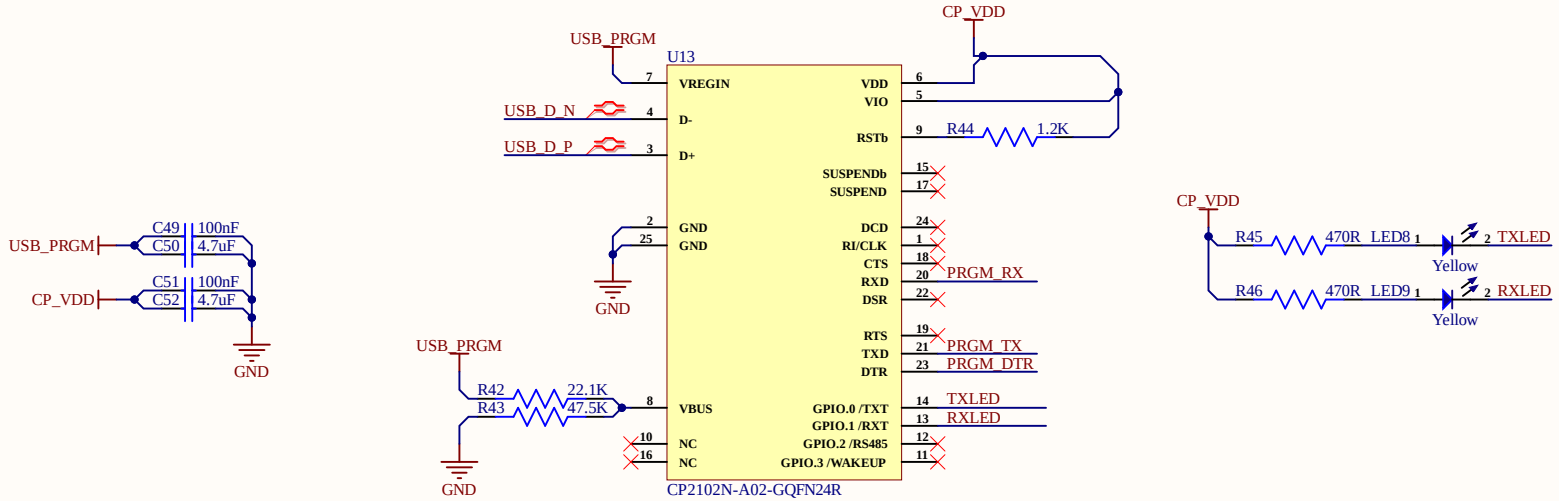
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GSM_UART
GSM_TX
GSM_RX
GSM_USB
GSM_USB_D_N
GSM_USB_D_P
PWRKEY_EN
GSM_RESET
USB_VBUS
GSM_PWR
VDD_EXT
GND

Title: GSM-Sheet.SchDoc	
File: Mega2560_Custom_v2.1.PrjPcb	
Date: 11/1/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


ATMEGA2560
U12
ATMEGA2560-16AU
Analog Inputs
J11
HA0200500000G
U14
TLV9301IDCKR
U15
TLV9301IDCKR
Fc = 106Hz
J12
BH254V-10P (JTAG 2.54mm)
J13
ICSP
SB1 is for Power through ICSP
SW1
X1
X322516MLB4SI
OLED10
OLED7
DNP
LED6
Yellow
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File: Mega2560_Custom_v2.1.PrjPcb
Date: 11/1/2023
Page Num: 6
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Page Total: 8

USB-to-UART for Programming





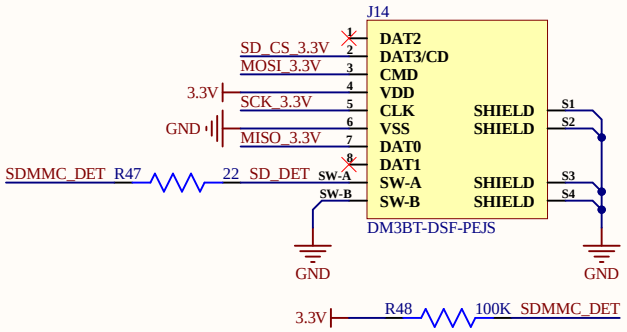


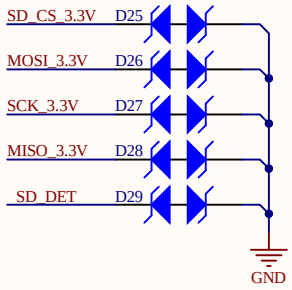


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Drawn By: Huzaifa Ali	Page Total: 8

SD Card Interface





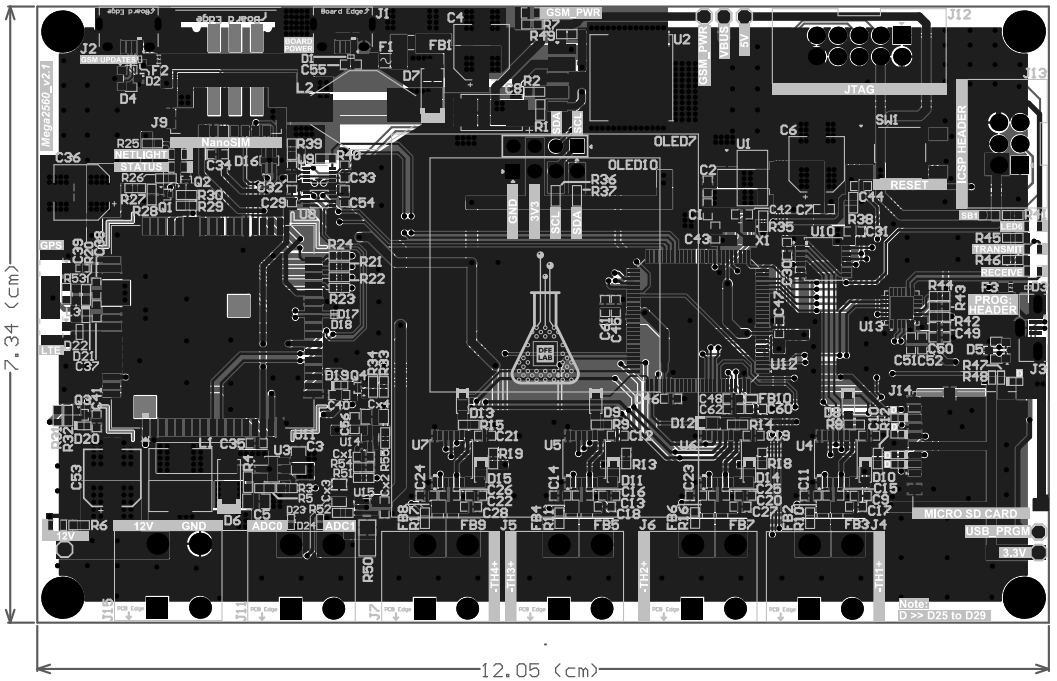






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Title: SDCard-Sheet.SchDoc	
File: Mega2560_Custom_v2.1.PrjPcb	
Date: 11/1/2023	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 8



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				
Total board thickness:			63.00mil		